

Integrated Visualization, Topology and Engineering Views

X.1 Executive Summary

The platform shall provide a unified, multi-scale visualization and topology intelligence capability covering geospatial, topology, single line diagram and substation engineering views. All views shall be driven by a common topology model, synchronized through CIM-compliant mRID identifiers, and governed through role-based controls, QA validation, lifecycle state management and auditable human approval workflows.

No.	View	Purpose
1	Geospatial View	Provides geographic context and spatial navigation using MapLibre.
2	Topology View	Provides logical network graph inspection using Cytoscape.js.
3	Single Line Diagram View	Provides schematic electrical hierarchy and feeder-level abstraction.
4	Substation Engineering View	Provides terminal-level connectivity, switching configuration validation and engineering-grade inspection.

The following platform elements shall support all visualization views:

No.	Platform Element	Required Role
1	Memgraph	Operational topology graph for runtime traversal and graph analytics.
2	PostGIS	Spatial geometry, staging and geospatial attribute layer.
3	NetworkX	QA and validation computation layer.
4	Kafka	Event propagation and synchronization across views and services.
5	CIM-compliant mRID	Persistent identity key across assets, terminals, connectivity nodes and topological relationships.

X.2 Visualization Hierarchy

No.	Level	View	Purpose
1	Macro	MapLibre	Geographic context.
2	Meso	Cytoscape	Network topology.
3	Schematic	SLD	Electrical hierarchy.
4	Micro	Substation Engineering View	Terminal-level connectivity and engineering inspection.

X.3 Core Design Principles

No.	ID	Principle	Requirement
1	VIS-01	Graph-first runtime	The system shall use Memgraph as the authoritative runtime topology graph for connectivity traversal, topology queries and operational graph analytics.
2	VIS-02	Separation of concerns	The system shall maintain clear boundaries between PostGIS for spatial geometry, Memgraph for topology, NetworkX for validation computation and Kafka for event propagation.
3	VIS-03	Unified identity	The system shall use CIM-compliant mRID identifiers as persistent keys across all visualization layers and integration services.
4	VIS-04	Consistent visual encoding	The system shall apply voltage, lifecycle, QA and operational states consistently across Map, Graph, SLD and Substation Engineering View.
5	VIS-05	Progressive detail	The user shall be able to navigate from geospatial overview to topology graph, SLD abstraction and substation-level engineering detail without losing asset context.

X.4 Single Line Diagram Capability

ID	No.	SLD Requirement
SLD-01	1	The SLD shall be generated from the operational topology graph in Memgraph.
SLD-02	2	The SLD shall present feeder and source-to-load hierarchy using a schematic layout.

SLD-03	3	The SLD shall support operational analysis, planning and switching scenario review.
SLD-04	4	The SLD shall remain synchronized with the geospatial and topology views through mRID-based identity mapping.

X.5 Substation Engineering View

X.5.1 Purpose

The system shall provide a detailed substation engineering view to support inspection of terminal-level connectivity, validation of switching configurations, controlled review of connection changes and evidence-based engineering assurance.

No.	Capability
1	Fine-grained inspection of busbar, breaker, disconnector, terminal and connectivity node relationships.
2	Verification of CIM Terminal and ConnectivityNode relationships.
3	Validation of operational TopologicalNode outcomes based on switch state.
4	Controlled proposal, simulation and approval of connection changes without bypassing governance controls.

X.5.2 Functional Scope

The substation view shall display:

No.	Component	Description
1	Busbars	Sectionalized and coupled arrangements.
2	Breakers	Open and closed switching state.
3	Disconnectors	Isolation points.
4	Terminals	CIM-level connection points.
5	Connectivity Nodes	Logical connection structure.
6	Bays	Optional equipment grouping.

X.5.3 Generation Model

The generation model shall follow a controlled pipeline from substation identity resolution to rendering.

No.	Generation Step
1	Resolve the selected substation mRID.
2	Query Memgraph for terminal-level equipment, terminal, connectivity node and switch-state relationships.
3	Normalize results against the CIM topology model, distinguishing Terminal, ConnectivityNode and TopologicalNode semantics.
4	Apply schematic layout rules for busbars, feeders, switching devices and parallel paths.
5	Render the Substation Engineering View and publish synchronization events for other visualization layers.

X.5.4 Source-of-Truth Boundaries

Layer	Authoritative Role	Primary Technology
Spatial geometry and staging	Stores coordinates, asset geometry, spatial attributes and GIS staging data.	PostGIS
Operational topology graph	Stores graph relationships used for traversal, connectivity, SLD generation and substation engineering view rendering.	Memgraph
Validation and QA computation	Executes topology checks, graph consistency rules, orphan detection and validation algorithms.	NetworkX
Event propagation	Distributes validated topology, QA, lifecycle and synchronization events across services.	Kafka
Identity	Provides persistent cross-system identity for assets, terminals, connectivity nodes and topology relationships.	CIM-compliant mRID

X.5.5 Layout Rules

Unlike SLD:

No.	SLD Characteristic	Substation View Characteristic
1	Hierarchical	Dense schematic.
2	Simplified	Detailed.
3	Feeder-level	Terminal-level.

X.5.6 Layout Characteristics

No.	Layout Characteristic
1	Busbars aligned horizontally.
2	Feeders connected vertically.
3	Switching devices clearly separated.
4	Parallel paths preserved.
5	Line crossing minimized.

X.6 Visual Encoding

X.6.1 Voltage Encoding

No.	Voltage	Colour	Note
1	33kV	Blue	Primary high-voltage visual encoding.
2	11kV	Red	Red is reserved for 11kV network visualization.
3	LV	Black	Low-voltage visual encoding.

X.6.2 Installation Encoding

No.	Installation Type	Line Style
1	Overhead	Solid line.
2	Underground	Dashed line.

X.6.3 Lifecycle Encoding

No.	Lifecycle State	Rendering
1	As-built	Full opacity.
2	Proposed	Reduced opacity.
3	Designed	Dot-dash.

X.6.4 QA and Operational Overlay

No.	QA / Operational State	Encoding
1	Error	Amber.

2	Processing	Yellow.
3	Pending approval	Purple.
4	Auto-fixed	Blue accent.
5	Rejected	Grey.

X.6.5 Substation-Specific States

No.	Component State	Visual Treatment
1	Closed breaker	Solid connection.
2	Open breaker	Gap or break symbol.
3	Isolated section	Faded or dashed.
4	Faulted component	Amber highlight.

X.7 Interaction and Synchronization

X.7.1 Unified Behaviour

All visualization views shall remain synchronized using mRID-based identity mapping and event-driven updates.

No.	User Action	Expected Result
1	Search POI	Map, Graph, SLD and Substation Engineering View are updated.
2	Select substation	Detailed substation view opens.
3	Select SLD node	Related asset is highlighted in the substation view.
4	Modify connection	Applicable views update through the approved workflow.

X.7.2 Engineering Interaction

Authorized users shall be able to inspect, analyze, simulate and propose engineering changes within the Substation Engineering View, subject to access control, QA validation and approval workflow requirements.

ID	No.	Interaction Type	Requirement
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INT-01	1	Inspection	Functions shall include breaker state review, terminal relationship inspection and connectivity node tracing.
INT-02	2	Analysis	Functions shall include power-flow path tracing, isolation-path identification and topology consistency checks.
INT-03	3	Simulation	Functions shall not directly execute field switching or production topology changes unless integrated with approved operational control procedures.
INT-04	4	Modification proposal	Proposed modifications shall be submitted to QA validation and maker-checker approval before promotion to the production topology graph.

X.8 Control and Navigation Tools

The system shall provide a controlled set of navigation, layer management, tracing, selection, engineering, search, history and safety tools to support efficient movement across Map, Graph, SLD and Substation Engineering View without loss of asset identity, lifecycle state, QA state or operational context.

X.8.1 View Navigation Tools

ID	No.	Tool / Function	Requirement
NAV-01	1	View controls	The system shall provide pan, zoom, fit-to-screen, reset-view and focus-on-selected-asset controls.
NAV-02	2	View switching	The system shall allow authorized users to switch between Map, Graph, SLD and Substation Engineering View while preserving selected asset mRID and view context.
NAV-03	3	Breadcrumb navigation	The system shall provide breadcrumb navigation from region to feeder, substation, bay and equipment level where supported by the data model.
NAV-04	4	Navigation history	The system shall provide back, forward and return-to-previous-context navigation for investigative workflows.

X.8.2 Layer Control Tools

ID	No.	Layer Control	Requirement
NAV-05	1	Voltage layers	The system shall allow users to toggle 33kV, 11kV and LV layers.

NAV-06	2	Installation type	The system shall allow users to toggle overhead and underground asset rendering.
NAV-07	3	Lifecycle state	The system shall allow users to toggle as-built, proposed and designed states.
NAV-08	4	QA and operational overlays	The system shall allow users to toggle QA overlays, operational states and switching states according to permissions.
NAV-09	5	Visual consistency	The system shall preserve approved visual encoding rules when layers are enabled, disabled or combined.

X.8.3 Topology Tracing Tools

ID	No.	Tracing Tool	Requirement
TRC-01	1	Upstream tracing	The system shall trace from a selected asset to the relevant source or upstream network component.
TRC-02	2	Downstream tracing	The system shall trace from a selected asset to connected downstream components.
TRC-03	3	Path and impact tracing	The system shall support feeder-path, isolation-boundary and affected-asset tracing where required topology and customer-impact data are available.
TRC-04	4	Trace evidence	Tracing results shall be visually distinguishable and shall preserve mRID-based trace evidence for audit and troubleshooting.

X.8.4 Measurement and Analysis Tools

The system shall provide measurement and analysis tools that support spatial, topology, operational-impact, QA and performance assessment without changing production topology or bypassing approved governance controls.

ID	No.	Measurement Tool	Requirement
MEAS-01	1	Distance measurement	The system shall measure line length, route length and feeder segment length where spatial geometry is available.
MEAS-02	2	Asset count measurement	The system shall count selected assets, transformers, poles, switches, terminals, connectivity nodes and graph elements within a selected view or boundary.

MEAS-03	3	Topology depth measurement	The system shall measure upstream and downstream topology depth from a selected asset by level, hop count or connected component where supported by graph data.
MEAS-04	4	Affected area measurement	The system shall estimate affected assets, feeders, substations, transformers, sections or customers within an isolation boundary where integrated data is available.
MEAS-05	5	Load and operational impact measurement	The system shall display estimated load, customer count, transformer count or other operational-impact measures where approved operational data is integrated.
MEAS-06	6	QA exception measurement	The system shall summarize QA exceptions by rule, severity, feeder, substation, owner, ageing and SLA status where QA workflow data is available.
MEAS-07	7	Performance measurement	The system shall capture or display render time, synchronization latency, query response time and QA queue refresh time for monitoring and test evidence where instrumentation is enabled.

X.8.5 Selection and Focus Tools

ID	No.	Selection / Focus Tool	Requirement
SEL-01	1	Select and focus	The system shall allow users to select an asset, focus the active view on the selected asset and highlight connected equipment.
SEL-02	2	Subnetwork investigation	The system shall support isolate-subnetwork, expand-neighbourhood and clear-selection actions.
SEL-03	3	Context preservation	The system shall maintain selected-asset context across synchronized views and display available metadata, QA state, lifecycle state and operational state.

X.8.6 Engineering Control Tools

ID	No.	Engineering Tool	Requirement
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ENG-01	1	Inspection	The system shall allow authorized users to inspect terminal connections, breaker states, disconnecter states and connectivity node relationships.
ENG-02	2	Switch simulation	The system shall allow authorized users to simulate switch open and close scenarios without directly executing production switching actions.
ENG-03	3	Change preview	The system shall allow users to preview proposed topology changes, compare current and proposed topology states, and submit proposed changes for QA validation and approval.
ENG-04	4	Mode clarity	The system shall clearly distinguish simulation mode, proposed-change mode and production view mode.

X.8.7 Search and Filter Tools

ID	No.	Search / Filter Tool	Requirement
SRCH-01	1	Search keys	The system shall support search by mRID, asset ID, substation name, feeder name, transformer, customer area and point of interest where data is available.
SRCH-02	2	Filters	The system shall support filtering by voltage, asset type, lifecycle state, QA state, operational state, installation type and stewardship assignment where applicable.
SRCH-03	3	Result navigation	Search and filter results shall support navigation to the relevant Map, Graph, SLD or Substation Engineering View.

X.8.8 History and Audit Navigation Tools

ID	No.	History / Audit Tool	Requirement
AUD-01	1	Change history	The system shall allow authorized users to view topology change history, prior approved states, rejected proposed changes and associated approval evidence.
AUD-02	2	Version comparison	The system shall support comparison of current and previous approved topology states where version history is available.

AUD-03	3	Audit context	Audit navigation shall display user, timestamp, asset mRID, change reason, workflow status and evidence reference for governed topology changes.
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X.8.9 Safety and Permission Controls

ID	No.	Safety / Permission Control	Requirement
SAFE-01	1	Restricted tools	The system shall disable restricted tools for unauthorized users and clearly indicate when a function is unavailable due to permissions.
SAFE-02	2	Production change protection	The system shall prevent direct production topology changes unless the approved QA, approval and promotion workflow has been completed.
SAFE-03	3	Mode labelling	The system shall provide clear labels for simulation mode, proposed-change mode and production mode to reduce operational risk.
SAFE-04	4	Audit logging	The system shall log restricted-tool attempts, simulations, proposed changes, approvals, rejections and promotions according to the approved audit policy.

X.9 QA, AI and Repair Integration

X.9.1 Substation-Level QA

The system shall detect and classify substation-level topology exceptions using machine-executable QA rules.

ID	No.	QA Exception Type
QAEX-01	1	Incorrect or missing terminal-to-connectivity-node relationships.
QAEX-02	2	Orphan nodes, orphan terminals and disconnected graph components.
QAEX-03	3	Invalid switch states or switching configurations.
QAEX-04	4	Topology outcomes that conflict with approved lifecycle or operational state.

X.9.2 AI-Assisted Repair Workflow

The AI-assisted repair workflow shall follow a human-governed sequence as defined below.

ID	No.	Workflow Step	Requirement
QA-01	1	Detect exception	The system shall identify topology or data-quality exceptions using approved QA rules.
QA-02	2	Analyze likely cause	The system shall support analysis of the likely source of the detected exception.
QA-03	3	Recommend fix	The system shall generate or present a proposed repair recommendation for review.
QA-04	4	Visualize proposed change	The system shall show the proposed repair as an overlay or before-and-after comparison without changing production topology.
QA-05	5	Submit for approval	The system shall route the proposed repair through the required QA and maker-checker approval workflow.
QA-06	6	Apply approved change	The system shall permit production update only after required approval and promotion controls are satisfied.
QA-07	7	Record audit evidence	The system shall retain evidence of detection, recommendation, approval, rejection, promotion and closure actions.

X.9.3 Lifecycle Visualization

ID	No.	Workflow Stage	Visual Treatment
QALC-01	1	Issue detected	Amber.
QALC-02	2	AI processing	Yellow.
QALC-03	3	Proposed fix	Overlay preview.
QALC-04	4	Approved	Updated topology.
QALC-05	5	Rejected	Grey.

X.9.4 QA Workflow Visualization

The system shall provide a dedicated QA workflow visualization capability that allows authorized users to view, investigate, track, compare, approve, reject and audit QA exceptions and proposed topology repairs from detection through closure.

ID	No.	QA Workflow Function	Requirement
QAWF-01	1	Exception queue	The system shall show exception type, affected asset mRID, affected node, edge, terminal or connectivity node, severity, workflow status, owner, ageing and SLA status where available.
QAWF-02	2	Exception drill-down	The system shall allow users to drill down from a QA exception to the affected Map, Graph, SLD or Substation Engineering View context.
QAWF-03	3	Workflow timeline	The system shall show detected, triaged, AI-analyzed, proposed fix generated, submitted for review, approved, rejected, escalated, applied and closed states.
QAWF-04	4	Topology comparison	The system shall provide before-and-after topology comparison for proposed repairs and distinguish current, proposed and approved topology states.
QAWF-05	5	Repair overlay	The system shall visualize proposed repair overlays without promoting the repair to production topology until required QA and approval workflow is completed.
QAWF-06	6	Reviewer decision controls	The system shall provide approve, reject, request rework, escalate and close controls, subject to role-based permission and segregation-of-duties controls.
QAWF-07	7	Evidence display	The system shall display evidence references, validation results, reviewer comments, decision history, timestamps and audit status.
QAWF-08	8	SLA and ageing	The system shall provide SLA and ageing indicators to support prioritization of unresolved QA exceptions.
QAWF-09	9	Audit evidence	The system shall preserve audit evidence for workflow transitions, reviewer decisions, automated recommendations and approved topology updates.

X.10 Search Integration

X.10.1 POI-Driven Substation View

The search function shall resolve a point of interest to its mRID, determine the asset type and route the user to the appropriate visualization view.

ID	No.	POI Search Scenario	Requirement
POI-01	1	Selected asset is a substation	The system shall open the Substation Engineering View.
POI-02	2	Selected asset is not a substation	The system shall provide navigation to the electrically related substation, feeder, SLD node or map location as applicable.
POI-03	3	Search navigation	The system shall preserve asset identity, lifecycle state and QA state across all synchronized views.

X.10.2 Contextual Actions

ID	No.	Contextual Action
ACT-01	1	Open substation detail.
ACT-02	2	Show switching configuration.
ACT-03	3	Simulate isolation scenario.
ACT-04	4	Trace upstream and downstream connectivity.
ACT-05	5	View QA exceptions and proposed repair actions.

X.11 Governance and Control Requirements

The system shall enforce governance controls to ensure that visualization, QA, simulation and topology modification activities are authorized, traceable and auditable.

ID	No.	Governance Control
GOV-01	1	Role-based access control shall restrict viewing, simulation, modification, approval and promotion rights by authorized role.
GOV-02	2	Maker-checker approval shall be required before any proposed connection change is promoted to the production topology graph.
GOV-03	3	QA topology, staging topology and production topology shall remain logically separated.
GOV-04	4	All connection updates, simulations, approvals, rejections and promotions shall be logged with user, timestamp, asset mRID, change reason and evidence reference.
GOV-05	5	Rollback or version recovery shall be available for approved topology changes where supported by the topology data management process.

GOV-06	6	Segregation of duties shall prevent the same user from both proposing and approving a production topology change.
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X.12 Operational Mode Definitions

ID	No.	Mode	Purpose	Allowed Actions	Restrictions and Controls
MODE-01	1	Production View Mode	Displays the current approved operational topology and asset state.	View, search, trace, inspect and audit approved topology.	Production topology shall not be modified directly outside the approved change workflow.
MODE-02	2	Simulation Mode	Allows users to test switching, isolation and topology scenarios without changing production topology.	Simulate, compare, trace, export evidence and discard scenario results.	Simulation outputs shall be clearly labelled and shall not execute field switching or production topology updates.
MODE-03	3	Proposed-Change Mode	Allows controlled preparation of topology changes for QA review and approval.	Create proposal, preview change, compare current and proposed state, submit for QA and approval.	Promotion to production shall require QA validation, maker-checker approval, segregation of duties and audit logging.
MODE-04	4	QA Review Mode	Allows authorized reviewers to assess QA exceptions and proposed repairs.	Review validation evidence, approve, reject, request rework, escalate or close QA workflow items.	Reviewer decisions shall be permission-controlled, timestamped and retained as audit evidence.

X.13 Roles and Permissions Matrix

The Roles and Permissions Matrix provides the governance baseline for assigning, enforcing and reviewing user access across visualization, tracing, simulation, proposal, approval, promotion and audit functions. It translates the platform's least-privilege, maker-checker and segregation-of-duties principles into role-specific permissions that must be consistently enforced across the Map, Graph, SLD and Substation Engineering View. The matrix shall be used to configure access controls, support periodic access reviews and provide audit evidence that restricted topology and workflow actions are performed only by authorized roles.

N o.	Role	View	Trace / Analy ze	Simul ate	Propos e Chang e	Appr ove / Rejec t	Promot e to Product ion	Audi t Acce ss	Key Control
1	Viewer	Allow ed	Limite d	Not allowe d	Not allowed	Not allow ed	Not allowed	Not allow ed	Read-only access.
2	GIS Steward	Allow ed	Allow ed	Limite d	Allowe d for GIS and staging correcti ons	Not allow ed	Not allowed	Limit ed	Cannot approve or promote own correctio ns.
3	Network Engineer	Allow ed	Allow ed	Allowe d	Allowe d	Not allow ed	Not allowed	Limit ed	Can propose but cannot approve or promote.
4	QA Reviewer	Allow ed	Allow ed	Allowe d	Not allowed	Allow ed	Not allowed	Allow ed	Reviews evidence but cannot promote.
5	Approver	Allow ed	Allow ed	Allowe d	Not allowed	Allow ed	Allowed , subject	Allow ed	Approval and promotio

							to policy		n require audit logging.
6	System Administrator	Allowed	Technical only	Not allowed unless separately authorized	Not allowed	Not allowed	Technical enablement only	Allowed	Administrative access shall not bypass business approval.

Permission legend: Allowed = role may perform the function within approved workflow controls; Limited = role may perform a restricted subset of the function; Technical only = access is limited to technical support and diagnostics; Not allowed = role shall not perform the function.

X.14 Visualization-Relevant QA Rule References

No.	Reference ID	QA Rule Category	Visualization Requirement	Primary Owning Domain
1	QA-VIS-01	Duplicate identity	The visualization layer shall flag duplicate mRID conditions where they affect asset selection, view synchronization or audit traceability.	Data Governance / Data Quality
2	QA-VIS-02	Missing terminal relationship	The visualization layer shall display missing terminal exceptions and allow drill-down to the affected equipment or connection point.	Data Governance / Topology QA
3	QA-VIS-03	Invalid connectivity node	The visualization layer shall highlight affected terminals and connectivity nodes using the approved QA overlay scheme.	Topology QA
4	QA-VIS-04	Disconnected graph component	The visualization layer shall show orphan nodes, disconnected components and broken feeder paths in context.	Topology QA / Network Engineering

5	QA-VIS-05	Invalid switch state	The visualization layer shall distinguish invalid switch-state outcomes and support escalation to engineering review.	Network Operations / Topology QA
6	QA-VIS-06	Lifecycle conflict	The visualization layer shall show lifecycle conflicts without promoting proposed or designed assets into production view mode.	Asset Data Governance
7	QA-VIS-07	Topology and geometry mismatch	The visualization layer shall display topology-geometry mismatch indicators and route the user to the relevant stewardship workflow.	GIS Governance / Topology QA

X.15 Visualization Synchronization Requirements

ID	No.	Synchronization Scenario	Visualization Requirement	Primary Owning Domain
SYNC-01	1	Asset selected in any view	The system shall synchronize the selected mRID across Map, Graph, SLD and Substation Engineering View while preserving user context.	Visualization / User Experience
SYNC-02	2	Approved topology change available	The visualization layer shall refresh affected views and clearly indicate when the displayed topology has been updated.	Integration Architecture / Topology Management
SYNC-03	3	QA exception raised	The visualization layer shall update the QA overlay, exception queue and affected asset context when QA exceptions are available.	Data Quality / QA Workflow
SYNC-04	4	Repair proposal submitted	The visualization layer shall display proposed repair overlays and workflow status without promoting changes to production topology.	QA Workflow / Integration Architecture

SYNC-05	5	Lifecycle state changed	The visualization layer shall update lifecycle rendering, filters and legends according to the latest approved lifecycle state.	Asset Data Governance / Integration Architecture
SYNC-06	6	View synchronization requested	The system shall route the user to the relevant view and asset context without losing selected mRID, QA state, lifecycle state or operational state.	Visualization / User Experience

X.16 Visualization Degraded Mode and Error Handling

ID	No.	Failure Condition	Required Behaviour	User Notification	Operational Control
RES-01	1	Memgraph unavailable	Disable topology tracing, SLD generation and substation engineering rendering that depend on live graph queries.	Display topology service unavailable message.	Prevent topology modification and log service failure.
RES-02	2	PostGIS unavailable	Disable or degrade geospatial rendering and spatial staging functions.	Display geospatial service unavailable message.	Preserve graph inspection where available and log failure.
RES-03	3	Kafka delay or outage	Continue local view operation where safe but mark synchronization as delayed.	Display synchronization delayed indicator.	Queue events for retry and prevent unsafe promotion where synchronization evidence is incomplete.
RES-04	4	QA engine failure	Disable automated QA validation and AI repair recommendation functions.	Display QA validation unavailable message.	Block promotion to production until QA validation is restored or approved manual

					control is applied.
RES-05	5	View synchronization failure	Maintain current view and prevent misleading cross-view context.	Display view synchronization failed message.	Require user refresh or retry and log affected mRID and view context.
RES-06	6	Permission service failure	Deny restricted actions by default.	Display authorization service unavailable message.	Fail closed for simulation, proposal, approval and promotion functions.

X.17 Non-Functional Requirements

ID	Requirement Area	Requirement
NFR-01	Performance	The system shall render standard substation views within an agreed SLA and shall support performance testing for complex substations with high equipment density.
NFR-02	Availability	The visualization service shall be designed for operational availability consistent with approved platform service levels.
NFR-03	Security	The system shall enforce authenticated access, role-based authorization and least-privilege permissions for engineering and topology operations.
NFR-04	Auditability	The system shall retain auditable records for visualization-driven topology changes, simulations, approvals and rejections.
NFR-05	Usability	The user interface shall support progressive drill-down, clear state legends and consistent navigation across Map, Graph, SLD and Substation Engineering View.
NFR-06	Accessibility	Visual encodings shall not rely on colour alone and shall include labels, symbols or patterns where operationally necessary.
NFR-07	Scalability	The architecture shall support growth in assets, substations, graph relationships, event volumes and concurrent visualization sessions.

NFR-08	Event latency	Topology, lifecycle and QA state changes shall propagate across synchronized views within agreed operational latency thresholds.
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X.18 Acceptance Criteria

No.	ID	Requirement Trace	Acceptance Criterion	Verification Method	Evidence
1	AC-SUB-01	X.5.3, X.5.4	A substation engineering view is generated from Memgraph using the selected substation mRID.	System demonstration and query review	Rendered view, query log and test record
2	AC-SUB-02	X.5.1, X.9.1	Terminal, ConnectivityNode and TopologicalNode relationships are represented according to the approved CIM topology model.	Model inspection and sample-data validation	CIM mapping evidence and QA test output
3	AC-SUB-03	X.6.5, X.7.2	Open and closed breaker or switch states are visibly distinguishable and reflected in topology tracing results.	Functional test	Screenshot set and test result
4	AC-SUB-04	X.9.1, X.9.3, X.9.4	QA exceptions are visible at component level and use the approved QA overlay scheme, including amber for detected errors.	Scenario test	QA scenario results and visual evidence
5	AC-SUB-05	X.7.1, X.8, X.10.1	Selecting an asset in one view updates the related Map, Graph, SLD or Substation Engineering View	End-to-end navigation test	Test script and navigation evidence

			without losing mRID context.		
6	AC-SUB-06	X.7.2, X.8.6, X.11	Simulation and proposed modification workflows require QA validation and approval before production topology promotion.	Workflow test and access-control review	Approval log, audit trail and role matrix
7	AC-SUB-07	X.6.3	As-built, proposed and designed lifecycle states are visually distinguishable across applicable views.	Visual inspection and regression test	Screenshot set and test record
8	AC-SUB-08	X.11	The same user cannot both propose and approve a production topology change.	Negative access-control test	Access test result and audit log
9	AC-SUB-09	X.9.4	QA workflow visualization shows exception queue, affected asset context, workflow timeline, proposed repair overlay, reviewer decisions, SLA status and audit evidence for each QA exception.	Workflow demonstration and scenario test	QA workflow screenshot set, approval trail and scenario test result
10	AC-SUB-10	X.12	Operational modes are clearly labelled, enforce permitted actions and prevent production topology modification outside	Mode-switching and workflow-control test	Mode screenshots, workflow logs and test result

			the approved workflow.		
11	AC-SUB-11	X.13	Role-based permissions enforce viewing, tracing, simulation, proposal, approval, promotion and audit access according to the approved matrix.	Access-control test	Role matrix, test account results and audit log
12	AC-SUB-12	X.14	Visualization-relevant QA rule references show how duplicate identity, missing terminal, invalid connectivity node, disconnected component, invalid switch state, lifecycle conflict and topology-geometry mismatch conditions are displayed, highlighted or routed from the visualization layer.	Visualization QA scenario test	QA overlay evidence, exception drill-down screenshots and stewardship routing evidence
13	AC-SUB-13	X.15	Visualization synchronization scenarios update affected views, QA overlays, lifecycle rendering and workflow status without losing mRID context.	View synchronization test	Synchronized view evidence, selected mRID trace and UI state screenshots
14	AC-SUB-14	X.8.4	Measurement and analysis tools calculate or display distance, asset counts, topology depth, affected area,	Measurement tool demonstration and scenario test	Measurement output screenshots, test data evidence and

			operational impact, QA exception summaries and performance metrics where supporting data and instrumentation are available.		instrumentation logs
15	AC-SUB-15	X.16	Failure of visualization dependencies such as Memgraph, PostGIS, Kafka, QA engine, view synchronization or permission service triggers the defined visualization degraded-mode behaviour and fails closed for restricted actions.	Visualization resilience and negative-path test	Failure-injection results, user notification screenshots and visualization audit logs
16	AC-SUB-16	X.17	Performance, security, auditability, usability, accessibility, scalability and event-latency requirements are validated against agreed test thresholds.	Non-functional test review	NFR test report

X.19 User Acceptance Testing

User Acceptance Testing shall confirm that business, engineering, GIS, QA, operations and support users can perform their assigned visualization, topology, measurement, QA workflow and governance tasks in a controlled test environment before operational sign-off.

X.19.1 UAT Participant Groups

No.	Participant Group	UAT Responsibility
1	GIS Steward	Validate geospatial navigation, asset selection, spatial measurements, GIS-related QA overlays and stewardship routing.
2	Network Engineer	Validate topology tracing, substation engineering view, switch simulation, measurement outputs and proposed-change workflow.
3	QA Reviewer	Validate QA exception queue, drill-down, workflow timeline, proposed repair overlays, review decisions and audit evidence.
4	Approver	Validate approval, rejection, promotion controls, segregation of duties and sign-off evidence.
5	Viewer / Operations User	Validate read-only navigation, search, view synchronization, contextual actions and operational visibility.
6	System Administrator	Validate technical support access, degraded-mode messages, permission enforcement and audit logging without bypassing business approval.

X.19.2 UAT Scenario Catalogue

No.	Scenario ID	Scenario	Primary Role	Expected Outcome
1	UAT-01	Select an asset on the map and open the related SLD.	Viewer / Operations User	The selected asset context and mRID are preserved across Map and SLD.
2	UAT-02	Search for a point of interest and open the Substation Engineering View.	GIS Steward	The correct substation view opens and displays the relevant asset context.
3	UAT-03	Trace upstream and downstream feeder path from a selected asset.	Network Engineer	Trace results are visually distinguishable and preserve trace evidence.
4	UAT-04	Run measurement tools for distance, asset count and affected area.	Network Engineer	Measurement outputs are displayed with

				supporting evidence where data is available.
5	UAT-05	Detect a QA exception and drill down to the affected node, edge, terminal or connectivity node.	QA Reviewer	The exception opens in the correct visualization context with QA state visible.
6	UAT-06	Review a proposed repair overlay and compare current versus proposed topology.	QA Reviewer	The proposed repair is visible without changing production topology.
7	UAT-07	Reject or approve a QA workflow item.	Approver	The decision is recorded with user, timestamp, evidence and workflow status.
8	UAT-08	Verify that an unauthorized user cannot promote a topology change.	System Administrator / QA Reviewer	The action is blocked, logged and clearly explained to the user.
9	UAT-09	Simulate switch open or close without affecting production topology.	Network Engineer	Simulation mode is clearly labelled and no production change occurs.
10	UAT-10	Confirm degraded-mode messages when a visualization dependency is unavailable.	System Administrator	The system displays the correct message and fails closed for restricted actions.

X.19.3 UAT Sign-Off Table

No .	Scenario ID	Role	Preconditions	Expected Outcome	Evidence	Pass / Fail	Sign-Off Owner
1	UAT-01 to UAT-10	Relevant UAT participant group	Test environment , test data, configured roles and	Scenario outcome matches	Screenshot, test log, audit log, measurement output or	To be completed during UAT.	Assigned business or technician

			baseline test scripts are available.	the expected result.	workflow evidence as applicable.		al owner.
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X.19.4 UAT Entry and Exit Criteria

No.	Criteria Type	Requirement
1	Entry	UAT environment is available, stable and configured with agreed test data.
2	Entry	UAT roles, permissions, scenarios, acceptance criteria and evidence templates are approved.
3	Entry	Critical system defects blocking UAT execution are resolved or formally waived.
4	Exit	Critical UAT scenarios are passed or formally accepted with documented exceptions.
5	Exit	High-severity defects are closed, mitigated or formally waived by the accountable owner.
6	Exit	Required UAT evidence is captured, reviewed and retained for audit and sign-off.
7	Exit	Business, engineering, GIS, QA, operations and support owners provide sign-off or documented conditional approval.

X.20 Traceability Matrix

No.	Requirement Area	Primary Requirement IDs	Primary Section	Acceptance Criteria	Verification Focus
1	Substation engineering generation	SUB-01, SUB-02, SLD-01	X.5.3, X.5.4	AC-SUB-01	Rendered view, graph query evidence and mRID resolution.
2	CIM topology representation	VIS-03, QAEX-01 to QAEX-04	X.5.1, X.9.1	AC-SUB-02	Terminal, ConnectivityNode and TopologicalNode model evidence.

3	Switching state visualization	INT-01, INT-02, TRC-01, TRC-02	X.6.5, X.7.2	AC-SUB-03	Breaker/switch state rendering and topology tracing.
4	QA visualization and workflow	QA-01 to QA-07, QALC-01 to QALC-05, QAWF-01 to QAWF-09	X.9.1, X.9.3, X.9.4	AC-SUB-04, AC-SUB-09	QA overlay, workflow timeline, proposed repair and audit evidence.
5	View navigation and synchronization	NAV-01 to NAV-09, SEL-01 to SEL-03, SRCH-01 to SRCH-03, POI-01 to POI-03, SYNC-01 to SYNC-06, SLD-04	X.7.1, X.8, X.10.1, X.15	AC-SUB-05, AC-SUB-13	Cross-view mRID preservation and synchronized UI state.
6	Measurement and analysis tools	MEAS-01 to MEAS-07	X.8.4	AC-SUB-14	Distance, counts, topology depth, affected area, operational impact, QA exception summary and performance metrics.
7	Governed engineering changes	ENG-01 to ENG-04, MODE-01 to MODE-04, SAFE-01 to SAFE-04, GOV-01 to GOV-06, AUD-01 to AUD-03	X.7.2, X.8.5, X.11, X.12	AC-SUB-06, AC-SUB-10	Simulation, proposal, QA validation, approval and promotion controls.
8	Lifecycle visualization	VIS-04, NAV-07, QALC-01 to QALC-05	X.6.3	AC-SUB-07	As-built, proposed and designed state rendering.
9	Segregation of duties and RBAC	GOV-01, GOV-02, GOV-06, SAFE-01 to SAFE-04	X.11, X.13	AC-SUB-08, AC-SUB-11	Access-control testing, role matrix evidence and audit logs.

10	Visualization QA rule references	QA-VIS-01 to QA-VIS-07	X.14	AC-SUB-12	Exception display, drill-down and stewardship routing.
11	Visualization degraded mode	RES-01 to RES-06	X.16	AC-SUB-15	Failure notifications, restricted-action fail-closed behaviour and logs.
12	Non-functional requirements	NFR-01 to NFR-08	X.17	AC-SUB-16	NFR test report and agreed thresholds.

X.21 Final Integrated Position

The platform shall support end-to-end grid visualization and control across four coordinated levels: geospatial map, network topology graph, single line diagram and substation engineering view.

The platform delivers a graph-native, CIM-compliant, multi-scale visualization framework that enables operational monitoring, topology validation, engineering-grade connection management and auditable human-governed repair workflows across the grid.

Appendix A: Cursor Implementation Prompt

Use the following prompt in Cursor to implement the Integrated Visualization, Topology and Engineering Views requirements as a production-grade, modular, testable feature set.

A.1 Cursor Prompt

You are a senior full-stack engineer and solution architect implementing an Integrated Visualization, Topology and Engineering Views capability for a grid utility platform. Implement the requirements in this document as a production-grade, modular, secure, auditable and testable feature set.

A.2 Product Goal

Build a unified visualization module providing four synchronized views: Geospatial View using MapLibre GL, Topology Graph View using Cytoscape.js, Single Line Diagram View generated from the operational topology graph, and Substation Engineering View for terminal-level inspection. All views must be synchronized using CIM-compliant mRID identifiers and must support role-based permissions, QA overlays, lifecycle states,

simulation mode, proposed-change workflows, maker-checker approval, audit logging and degraded-mode behavior.

A.3 Technology Boundaries

- PostGIS shall remain the source of truth for spatial geometry, GIS staging, coordinates, asset geometry and spatial attributes.
- Memgraph shall remain the authoritative runtime topology graph for traversal, SLD generation, topology relationships and substation engineering rendering.
- NetworkX shall provide QA and validation computation, graph consistency checks, orphan detection and topology validation.
- Kafka shall provide event propagation and cross-view synchronization.
- CIM mRID shall be the persistent identity key across assets, terminals, connectivity nodes, topological nodes and relationships.
- MapLibre GL shall render the geospatial view, and Cytoscape.js shall render graph-based topology and engineering views.

A.4 Operational Boundaries

- Production topology shall remain protected and must not be modified directly from visualization, SLD, simulation or substation engineering views.
- Simulation mode shall be read-only against production and must not execute field switching, dispatch actions or production topology updates.
- Proposed-change mode shall create controlled proposals only; promotion to production shall require QA validation, maker-checker approval, segregation of duties and audit logging.
- QA review mode shall allow authorized review, approval, rejection, rework, escalation and closure decisions, but only within role-based and auditable workflow controls.
- QA topology, staging topology, simulation state and production topology shall remain logically separated.
- Operational control procedures shall remain authoritative for live switching and field operations; visualization tools shall support analysis and evidence, not bypass approved operational control.
- If synchronization, QA validation, permission checks or dependency evidence is incomplete, restricted actions shall fail closed and production promotion shall be blocked.

A.5 Required Implementation Scope

1. Inspect the existing codebase structure and identify correct integration points.

2. Add or align domain types, constants, visual encodings, permissions, operational modes and QA workflow types.
3. Implement permission and mode enforcement before adding write or proposal actions.
4. Implement selected mRID state management and cross-view synchronization.
5. Implement Map, Graph, SLD and Substation Engineering View shells.
6. Implement QA workflow panel, exception queue, workflow timeline, proposed repair overlays and reviewer decision controls.
7. Implement search, filter, layer control, navigation, measurement and audit-history panels.
8. Implement audit logging and degraded-mode fail-closed behavior.
9. Add automated tests for synchronization, permissions, mode restrictions, QA workflow, degraded mode, measurement, audit events and accessibility of visual encodings.
10. Run lint, type checks and tests, then produce a concise implementation summary listing files changed, requirements implemented, tests added and known gaps or assumptions.

A.6 Domain Model Requirements

Define strong types for mRID, Asset, Substation, Feeder, Terminal, ConnectivityNode, TopologicalNode, SwitchingDevice, Breaker, Disconnecter, TopologyEdge, TopologyGraph, LifecycleState, QAState, OperationalState, VisualizationView, OperationalMode, UserRole, Permission, QAException, RepairProposal, WorkflowStatus, AuditEvent, MeasurementResult and DegradedModeCondition. Use mRID as the cross-view identity key everywhere.

A.7 View Requirements

- Geospatial View: support pan, zoom, fit-to-screen, reset-view, selected-asset focus, voltage layers, installation layers, lifecycle overlays, QA overlays, operational overlays, asset selection and search navigation.
- Topology Graph View: support upstream, downstream, feeder-path, isolation-boundary and affected-asset tracing; selection by mRID; QA overlays; disconnected component and orphan-node distinction.
- Single Line Diagram View: generate schematic source-to-load hierarchy from Memgraph, synchronize mRID selection and support operational analysis without direct production topology change.

- Substation Engineering View: render busbars, breakers, disconnectors, terminals, connectivity nodes, bays where available, terminal-to-connectivity-node relationships, connectivity-node-to-topological-node outcomes and switch states.

A.8 Visual Encoding and Accessibility

- 33kV shall be blue, 11kV shall be red and LV shall be black. Red is reserved for 11kV network visualization and must not be used for errors.
- Overhead assets shall use solid lines; underground assets shall use dashed lines.
- As-built shall use full opacity, proposed shall use reduced opacity and designed shall use dot-dash treatment.
- Detected issues shall use amber, processing shall use yellow, pending approval shall use purple, auto-fixed shall use blue accent and rejected shall use grey.
- Closed breakers shall show solid connection, open breakers shall show a gap or break symbol, isolated sections shall be faded or dashed and faulted components shall use amber highlight.
- Do not rely on colour alone. Include labels, symbols or patterns where needed for accessibility.

A.9 Governance, QA and Workflow Controls

- Implement production view mode, simulation mode, proposed-change mode and QA review mode with clear UI labels and enforced restrictions.
- Implement RBAC for Viewer, GIS Steward, Network Engineer, QA Reviewer, Approver and System Administrator roles according to the document's permission matrix.
- Enforce least privilege, segregation of duties and the rule that the same user must not both propose and approve a production topology change.
- Administrative access must not bypass business approval.
- Restricted actions must fail closed if the permission service fails.
- QA workflow must support exception detection, analysis, recommended fix, proposed repair overlay, submission for approval, approved application and audit evidence recording.

A.10 Degraded Mode and Audit Logging

Implement degraded-mode handling for Memgraph, PostGIS, Kafka, QA engine, synchronization and permission-service failures. Disable unsafe capabilities, show clear user notifications, queue safe retries where appropriate, block production promotion when evidence is incomplete and record audit events for failures, restricted-tool attempts, simulations, proposed changes, QA exceptions, repair recommendations, approvals, rejections, requests for rework, escalations, closures and promotions to production.

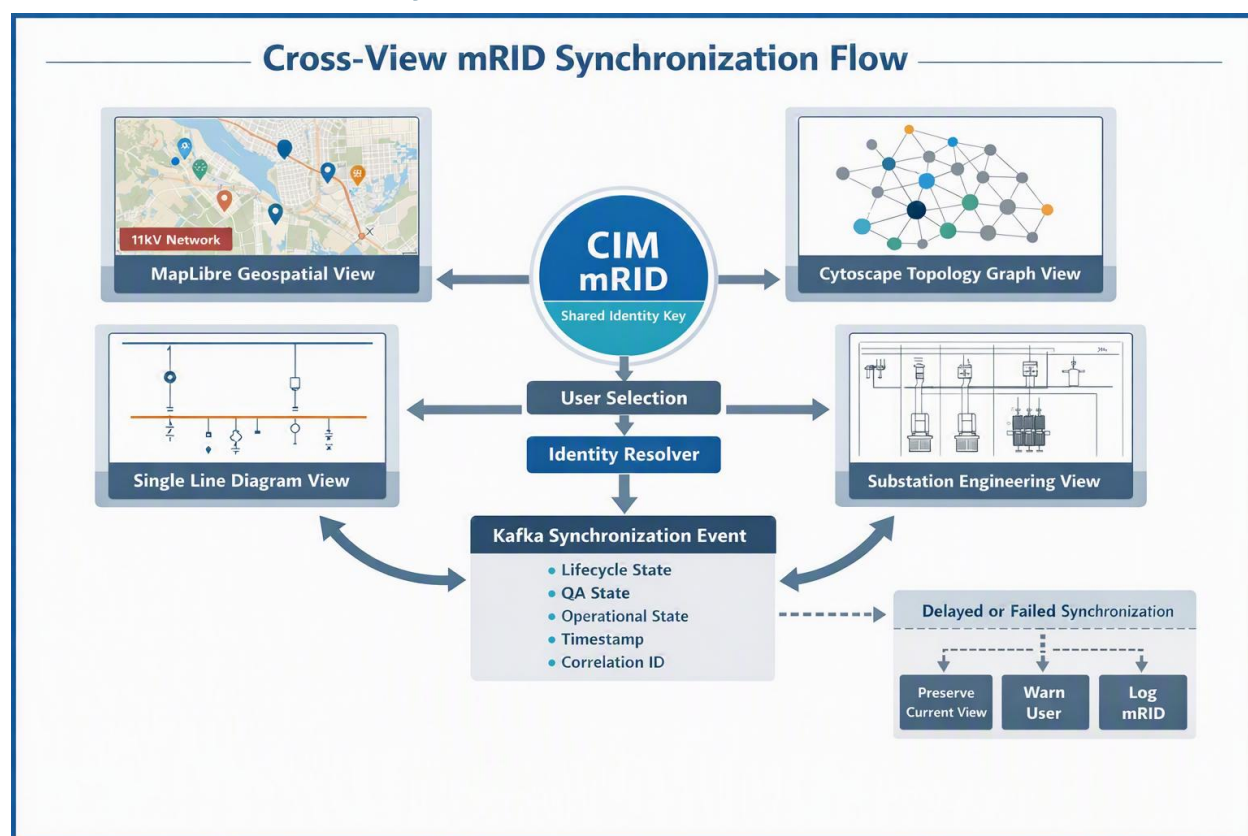
A.11 Acceptance Criteria

Implementation is complete only when the acceptance criteria in Section X.18 are satisfied, including Memgraph-based substation engineering rendering, CIM topology representation, breaker and switch-state visibility, QA overlays, cross-view mRID synchronization, governed simulation and proposed-change workflows, lifecycle rendering, segregation of duties, QA workflow visualization, mode enforcement, RBAC enforcement, degraded-mode behavior, measurement tools, instrumentation and non-functional validation.

Appendix B: Implementation Interaction Diagrams

This appendix provides text-based implementation diagrams to clarify synchronization, governance workflow and service-boundary behaviour for delivery teams and vendors.

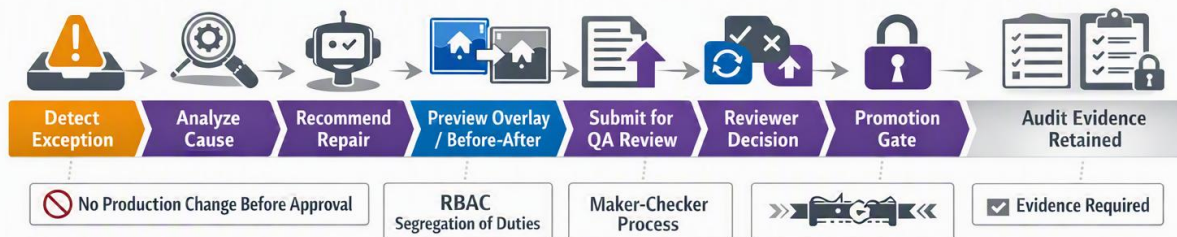
B.1 Cross-View mRID Synchronization Flow



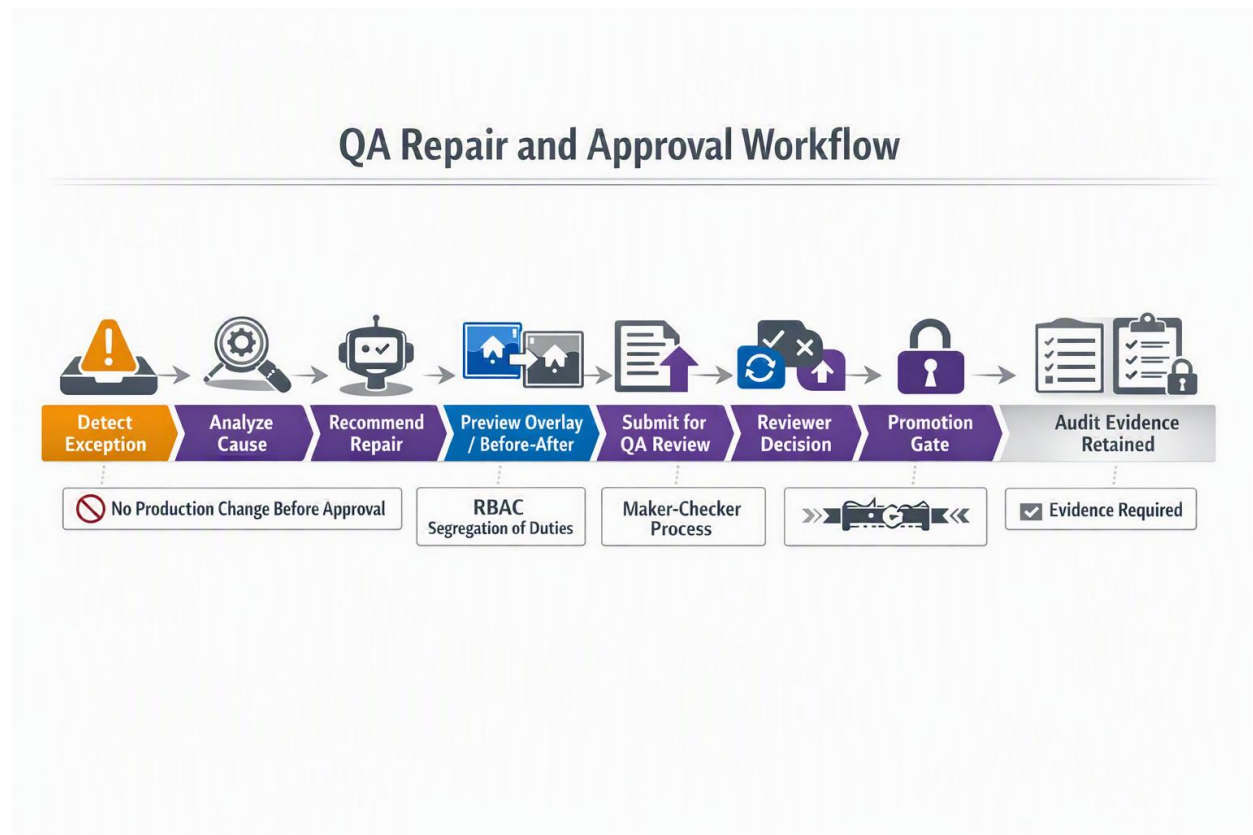
Step	Interaction	Required Behaviour
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1	User selects an asset in Map, Graph, SLD or Substation Engineering View.	The selected asset mRID becomes the authoritative UI context.
2	Identity resolver validates the selected mRID and asset type.	The system preserves asset identity, lifecycle state, QA state and operational state.
3	Synchronization service publishes a Kafka-backed selection event.	Event includes selected mRID, source view, target views, lifecycle state, QA state, operational state, timestamp and correlation ID.
4	Target views consume or receive the synchronization event.	Map, Graph, SLD and Substation Engineering View update without losing mRID context.
5	Applicable overlays refresh.	Lifecycle, QA, operational, trace and workflow overlays render consistently using approved visual encodings.
6	Synchronization delay or failure occurs.	System maintains current view, marks synchronization as delayed or failed, prevents misleading context and logs affected mRID and correlation ID.

QA Repair and Approval Workflow



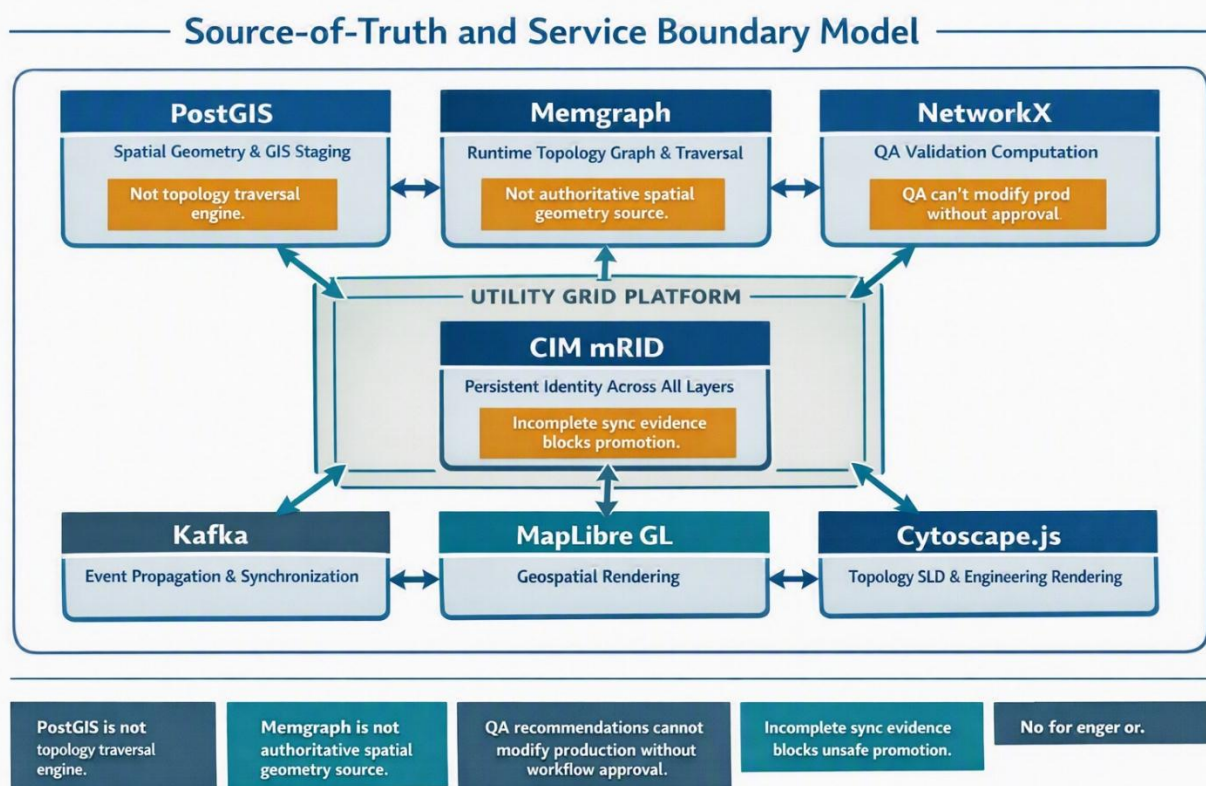
B.2 QA Repair and Approval Workflow



Step	Workflow Stage	Control Requirement
1	Detect exception.	QA engine or validation service identifies exception type, affected mRID, severity and affected component.
2	Analyze likely cause.	System links exception to topology, terminal, connectivity node, switch-state, lifecycle or geometry evidence.
3	Recommend repair.	AI-assisted or rules-based recommendation is generated without changing production topology.
4	Preview proposed change.	Visualization shows overlay or before-and-after comparison in proposed-change or QA review context.
5	Submit for QA review.	Workflow item records submitter, timestamp, affected mRID, evidence reference and correlation ID.
6	Reviewer decision.	Authorized reviewer may approve, reject, request rework, escalate or close according to RBAC and segregation-of-duties controls.

7	Promotion decision.	Production promotion is allowed only after QA validation, maker-checker approval, required evidence and permission checks are complete.
8	Audit evidence retained.	Audit log captures user, role, timestamp, action, asset mRID, workflow item ID, decision, evidence reference, source view, mode and correlation ID.

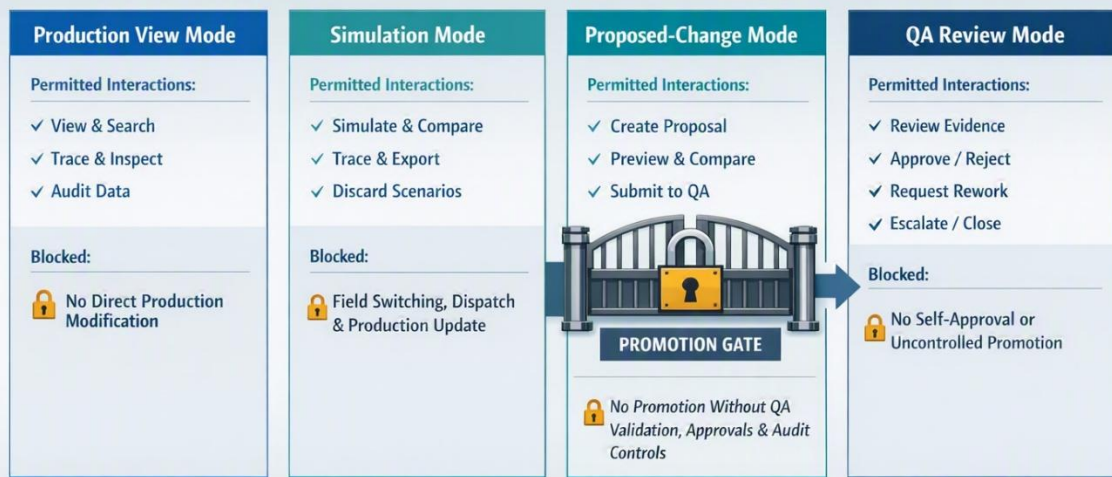
B.3 Source-of-Truth and Service Boundary Diagram



Layer / Service	Authoritative Responsibility	Boundary Rule
PostGIS	Spatial geometry, GIS staging, coordinates, asset geometry and spatial attributes.	Do not use PostGIS as the runtime topology traversal engine.
Memgraph	Runtime topology graph, topology traversal, SLD generation and substation engineering rendering.	Do not store spatial geometry in Memgraph as the authoritative spatial source.

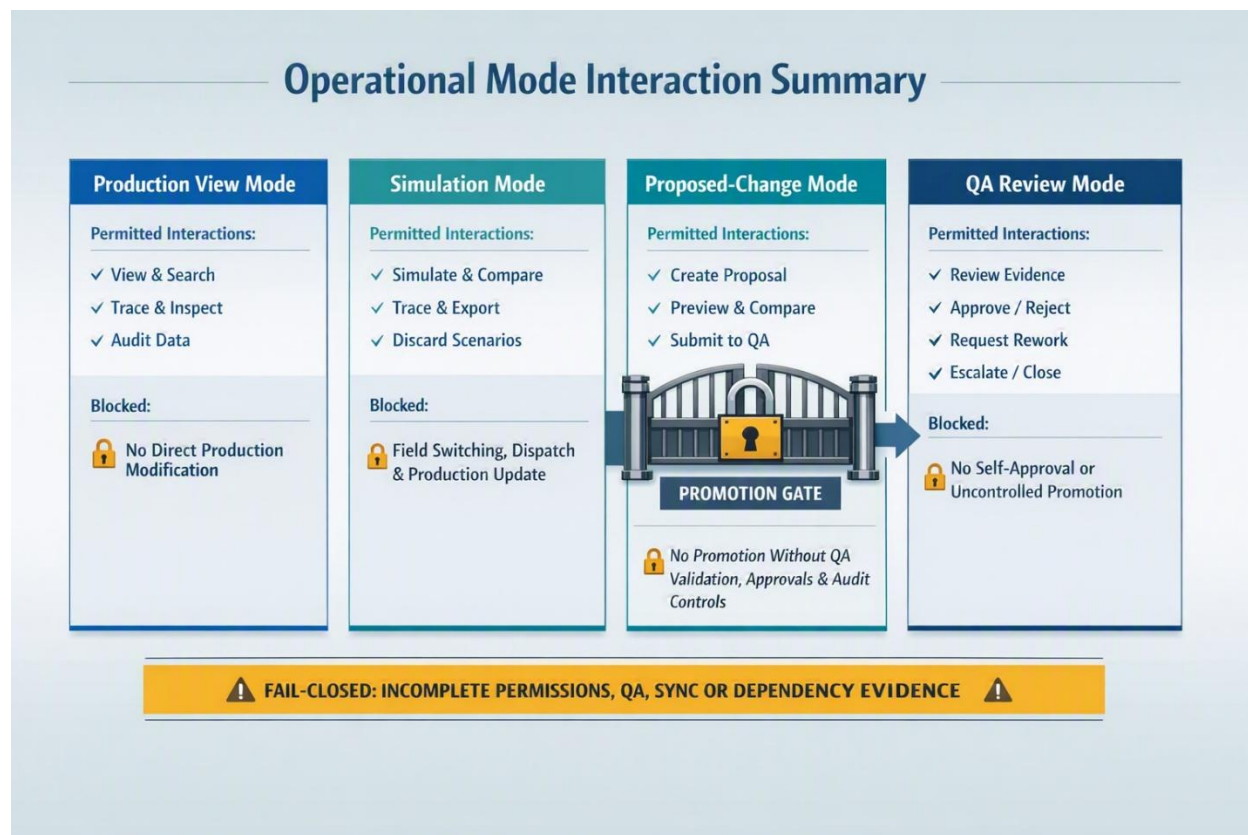
NetworkX	QA computation, graph consistency checks, orphan detection and validation algorithms.	Do not allow QA recommendations to modify production topology without approval workflow controls.
Kafka	Event propagation, synchronization and state-change distribution.	Delayed or incomplete synchronization evidence shall block unsafe promotion.
CIM mRID	Persistent identity across assets, terminals, connectivity nodes, topological nodes and relationships.	All cross-view selection, audit and workflow references shall preserve mRID context.
MapLibre GL	Geospatial rendering and map-based navigation.	Map rendering shall consume approved spatial data and synchronized mRID context.
Cytoscape.js	Topology, SLD and engineering graph rendering.	Graph rendering shall reflect Memgraph-derived topology and approved visual encodings.

Operational Mode Interaction Summary



⚠️ FAIL-CLOSED: INCOMPLETE PERMISSIONS, QA, SYNC OR DEPENDENCY EVIDENCE ⚠️

B.4 Operational Mode Interaction Summary



Mode	Permitted Interaction	Blocked Interaction
Production View Mode	View, search, trace, inspect and audit approved topology.	Direct production topology modification outside approved workflow.
Simulation Mode	Simulate, compare, trace, export evidence and discard scenarios.	Field switching, dispatch action or production topology update.
Proposed-Change Mode	Create proposal, preview change, compare current and proposed state and submit for QA review.	Production promotion without QA validation, approval, segregation of duties and audit evidence.
QA Review Mode	Review evidence, approve, reject, request rework, escalate or close workflow item.	Uncontrolled promotion, self-approval or bypass of permission controls.